

```
rq1_agreement <- lmer(wake_diff_booklet ~ wake_diff_cap + (1 | SubjectID), data = analysis_data)

# LMM

summary(rq1_agreement)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
```

```
## lmerModLmerTest]
```

```
## Formula: wake_diff_booklet ~ wake_diff_cap + (1 | SubjectID)
```

```
## Data: analysis_dat
```

```
##
```

```
## REML criterion at convergence: 2687.4
```

```
##
```

```
## Scaled residuals:
```

```
##      Min       1Q   Median       3Q      Max
```

```
## -5.8429  0.0151  0.1727  0.2957  4.3420
```

```
##
```

```
## Random effects:
```

```
## Groups      Name      Variance Std.Dev.
```

```
## SubjectID (Intercept)  37.34    6.11
```

```
## Residual                1022.32   31.97
```

```
## Number of obs: 274, groups: SubjectID, 30
```

```
##
```

```
## Fixed effects:
```

```
##              Estimate Std. Error      df t value Pr(>|t|)
```

```
## (Intercept)    -6.703497    2.972431   65.358396   -2.255    0.0275 *
## wake_diff_cap    0.993702    0.007309  253.113242  135.960   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##
##          (Intr)
## wake_dff_cp -0.652
```

```
car::linearHypothesis(rq1_agreement, "wake_diff_cap = 1")
```

```
##
## Linear hypothesis test:
## wake_diff_cap = 1
##
## Model 1: restricted model
## Model 2: wake_diff_booklet ~ wake_diff_cap + (1 | SubjectID)
##
##   Df  Chisq Pr(>Chisq)
## 1
## 2  1 0.7424    0.3889
```

```

# Test if slope is equal to 1

rq1_dat <- rq1_agreement@frame %>%

  dplyr::mutate(row_id = as.numeric(rownames(rq1_agreement@frame)))

# Extract data used in model

rq1_dat$pred_booklet <- predict(rq1_agreement, re.form = NA)

# Add predictions (population level)

rq1_plot <- ggplot(rq1_dat, aes(x = wake_diff_cap, y = wake_diff_booklet)) +

  geom_point(alpha = 0.6) +

  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed", linewidth =

  geom_line(aes(y = pred_booklet), color = "blue", linewidth = 1) +

  labs(

    x = "Cap Time Since Wake (min)",

    y = "Booklet Time Since Wake (min)",

    title = "Agreement Between Booklet and Cap"

  ) +

  facet_zoom(xlim = c(0, 50), ylim = c(0, 50), horizontal = FALSE)

# Zoom in from wake-50 min

```

TODO - update rq2 to be a nice table and insert table below

An Instigation into Time Agreement, Subject Adherence, and Patterns of Cortisol and DHEA in Novel Saliva Collection Device

Ezra Weible¹

Introduction

0.5 pages; describe data and scientific hypotheses of interest. Rephrase scientific hypotheses into statistical hypotheses.

Methods

2 pages; past tense; no results; brief description of data management and cleaning. In depth description of analysis methods. Justify and explain data analysis approach.

¹Biostatistics Masters Student, Colorado School of Public Health, University of Colorado Anschutz Medical Campus, Aurora, CO

Results

1-1.5 pages; present results from analysis. Also include descriptive statistics and model selection, if applicable. Reference tables and figures here. Full interpretation of results including pt estimates/CI/pvals/interpretation.

Discussion

0.5-1pg; interpret results in context of scientific Qs; discuss limitations

Appendix

Directory: Github

File name: P0/Code/P0_Report.Rmd

Tables and Figures:

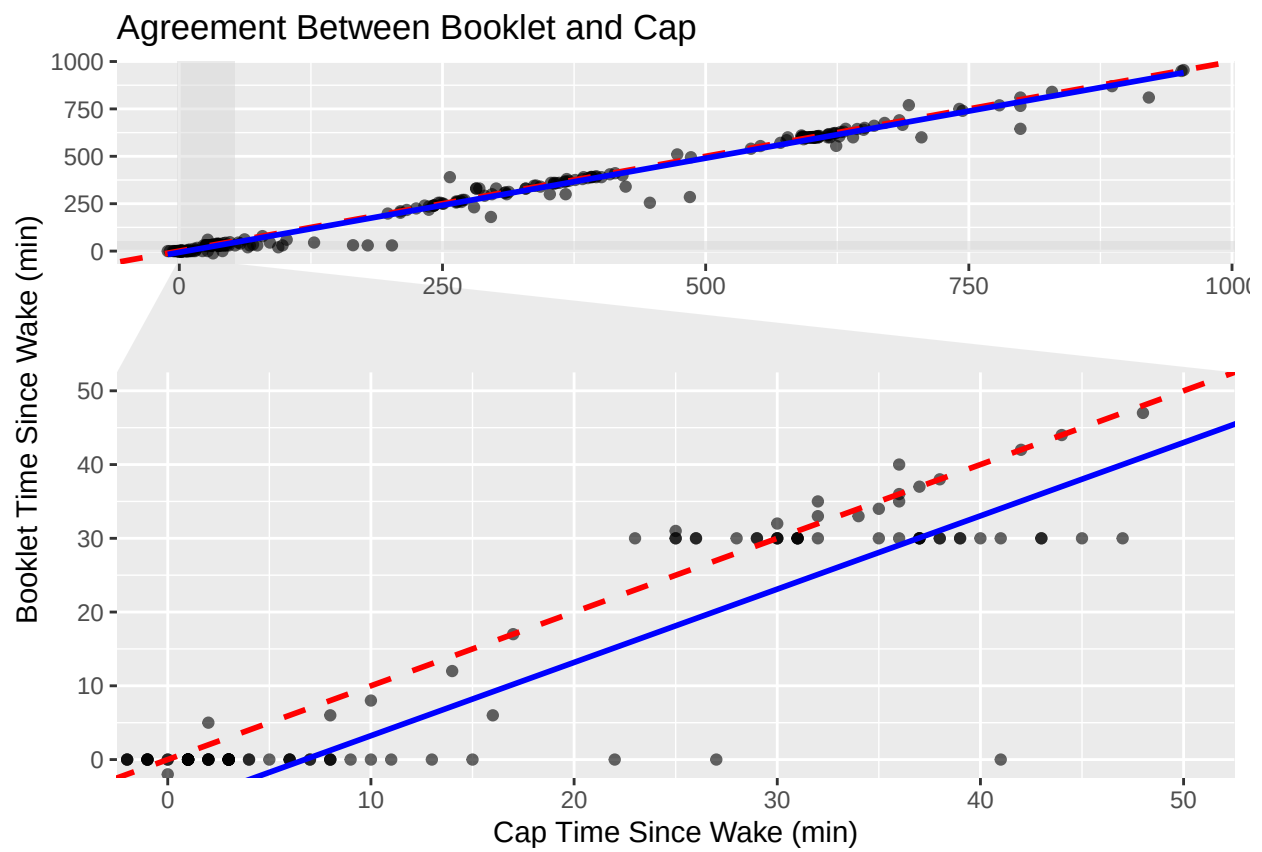


Figure 1: (Research Question 1) Agreement between reported booklet time since wake and reported cap time since wake.

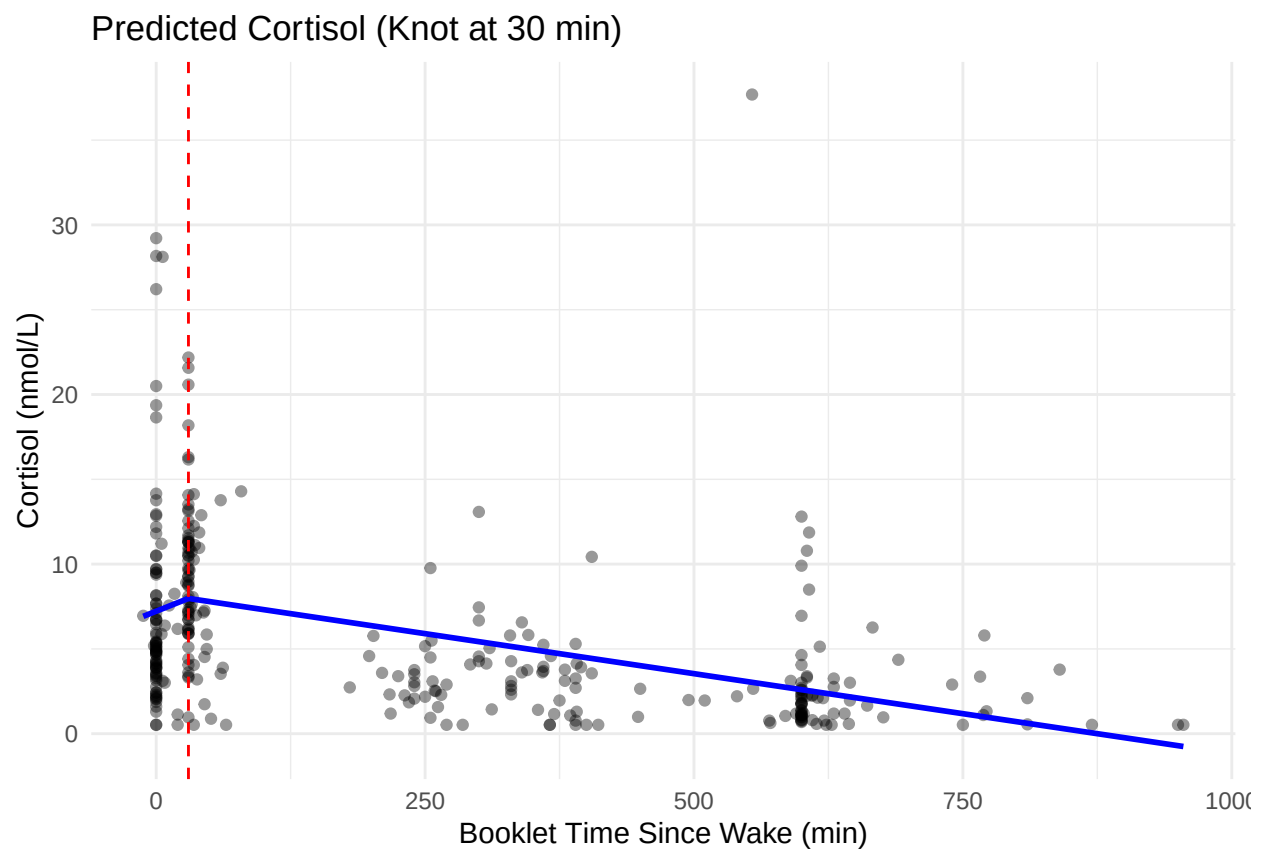


Figure 2: (Research Question 3) Predicted trend of cortisol over time since wake, reported from booklet.

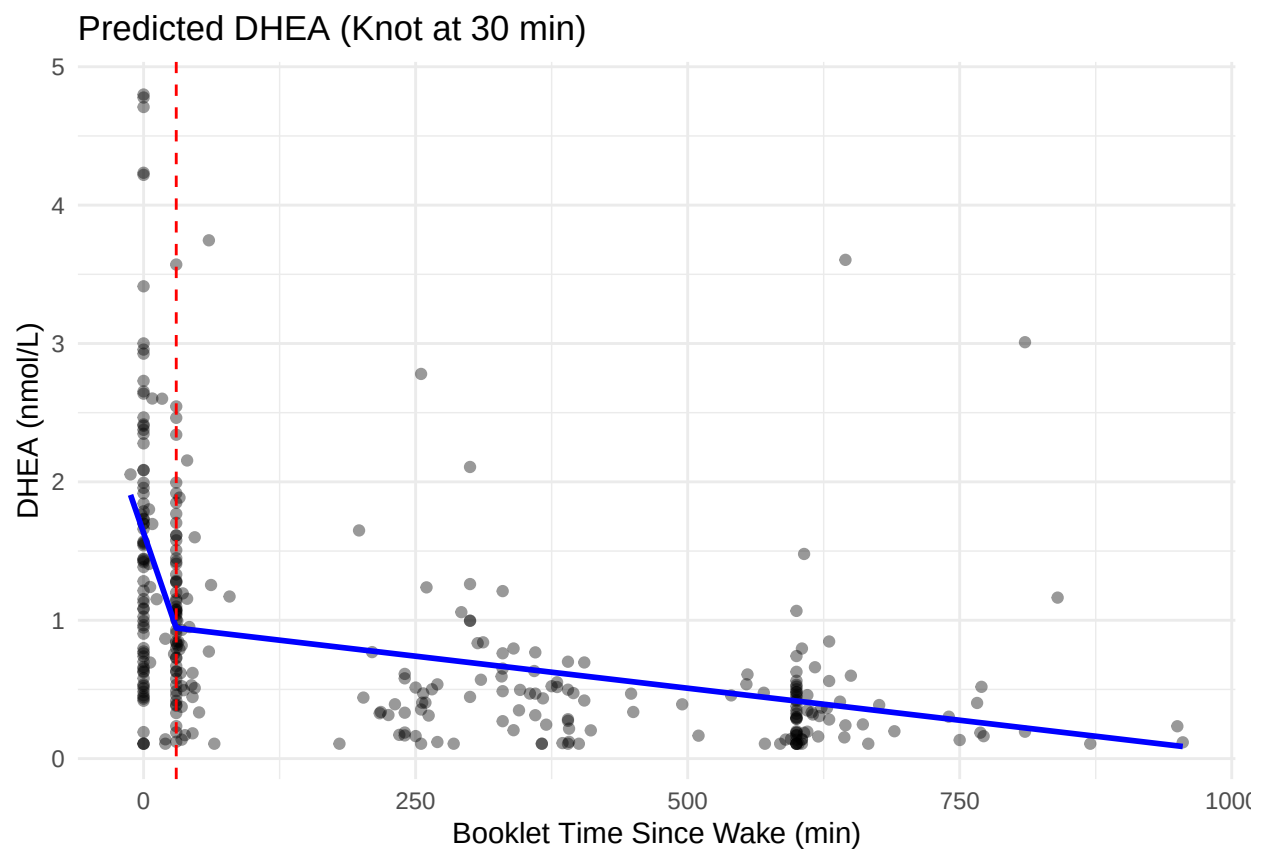


Figure 3: (Research Question 3) Predicted trend of DHEA over time since wake, reported from booklet.